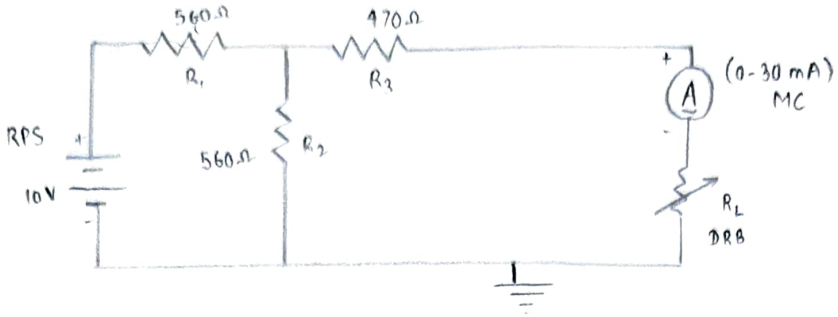
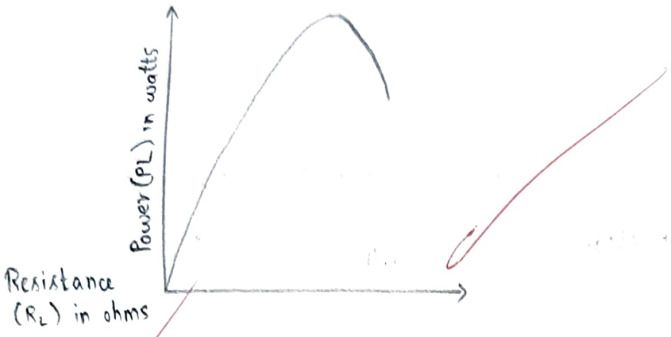
CIRCUIT DIAGRAM:



MODEL GRAPH:



TABULATION:

TABLE (1): FOR THEORETICAL CALCULATION : $R_{th} = 750\Omega$

S.No	Load Resistance (R_L) in Ohm	Load Current (I_L) in mA	Load Voltage (V_L) in Volts	Load Power (P_L) in Watts
1.	750	3.33	2.5	8.33×10^{-3}
2.	470	4.09	1.92	7.85×10^{-3}
3.	560	3.81	2.13	8.115×10^{-3}
4.	850	3.12	2.65	8.26×10^{-3}
5.	1kΩ	2.85	2.85	8.12×10^{-3}

TABLE (2): FOR PRACTICAL CALCULATION:

S.NO	Load Resistance (R_L) in Ohm	Load Current (I_L) in Amps	Load Voltage (V_L) in Volts	Load Power (P_L) in Watts
1.	470	4.09	1.92	7.85×10^{-3}
2.	560	3.81	2.13	8.115×10^{-3}
3.	750	3.33	2.5	8.3×10^{-3}
4.	850	3.12	2.65	8.26×10^{-3}
5.	1kΩ	2.85	2.85	8.12×10^{-3}

Incorrect

EXP. NO: 6
24/2/26

VERIFICATION OF MAXIMUM POWER TRANSFER THEOREM

AIM:

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

APPARATUS REQUIRED:

S.NO	Name of the apparatus	Range / Rating	Quantity
1.	Voltmeter	(0-15V) MC	1
2.	Ammeter	(0-500mA) MC	1
3.	Resistors	560Ω, 470Ω	2, 1
4.	RPS (DC Supply)	15V	1

PROCEDURE:

1. Make connection as per the circuit diagram.
2. Change the resistors R_L whose value close to R_{th}, measure the corresponding V_L, I_L and calculate the power P_L, and enter into the table (2).
3. Plot a graph between R_L and P_L and find the R_L corresponding to maximum power transfer.
4. Verify the measured values of R_L at maximum power transfer as same as calculated and found graphically.

MODEL CALCULATION:

$$V_{Th} = \frac{V \times R_2}{R_1 + R_2} = \frac{10 \times 560}{500} = 5V$$

$R_{Th} = R_1$ and R_2 is Parallel

$$R_{Th} = \frac{560 \times 560}{2 \times 560} + 470$$

$$= 280 + 470$$

$$R_{Th} = 750\Omega$$

$$R_L = R_{Th}$$

$$P_L = \frac{(V_{Th})^2}{4 \times R_{Th}} = \frac{5^2}{4 \times 750}$$

$$= \frac{25}{3000}$$

$$P_L = 0.0083\ mW$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{5}{1500} = 0.0033$$

$$I_L = 3.33\ mA$$

$$V_L = I_L \times R_L = 3.33 \times 10^{-3} \times 750$$

$$V_L = 2.5V$$

MARK SPLITUP:

SL.NO	Description	Marks Alloted	Marks Obtained
1.	Experimental Setup / Connections	20	20
2.	Observation / Readings	20	20
3.	Calculation / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total:	100

100

24/2/20

RESULT:

Thus, maximum power transfer theorem is verified practically and theoretically.